

Gap Register ↔ AI-Adjusted Timeline Crosswalk

Every gap in the Director-approved Aug–Sep plan, pinned against its matching task(s) in the 6-June AI-Adjusted Project Timeline — checked for effort agreement, dependency agreement, and whether a task exists at all.

SOURCE A

DA_gaps.pdf plan eval, 07 Aug 2026

SOURCE B

AI-Adjusted Timeline, 06 Jun 2026

GAPS CHECKED

14 / 14

PREPARED FOR

Chandra Sekhar — Backend Lead

01 — RESULT AT A GLANCE

Of the 14 gaps in the plan's Section 02 register, 11 map cleanly onto one or more AI-Adjusted Timeline tasks with matching scope and dependency order. Three do not — one has no timeline task at all, and two carry effort figures that read as inconsistent between the two documents once the "AI effort" and "PD" units are converted to the same basis.

11

Gaps aligned — task exists, scope and dependency chain match

2

Gaps to watch — task exists, effort basis diverges

1

Gap with no timeline task — G14 frontend test suite

02 — HOW TO READ A CARD

Each gap card below sets the plan's own numbers (Person-Days, dependency ref) against the timeline's matching task line (AI Effort symbol, note). The centre pin marks the verdict for that specific pairing.

- Aligned — same scope, same dependency, effort converts cleanly
- Watch — task exists but effort basis or scope needs a second look
- Mismatch — no corresponding timeline task found

03 — GAP-BY-GAP CROSSWALK

All 14 gaps, in the plan's original G1–G14 order. PD figures are the plan's own estimates; AI Effort figures and task IDs are quoted directly from the timeline document.

G1 3D visualisation scaffold (Cesium / terrain)

FRONTEND

ALIGNED

PLAN REGISTER (SOURCE A)

10 person-days · dep: none

Milestone target: 31 Aug, "usable scaffold."

TIMELINE (SOURCE B)

P3-01 + P3-02

XL (3–4 days) + L (1.5 days) $\approx$ 4.5–5.5 days

Timeline explicitly flags Cesium as "the dominant cost — a 600k-line library with its own paradigm." Both documents independently identify the same task as the single largest line item in their respective phase.

Match quality: Strong. Both sources isolate Cesium integration as the long pole. 10 PD (human-paced) vs 4.5–5.5 AI-adjusted days is consistent with the timeline's own stated 3–4× bottleneck multiplier for novel third-party libraries.

G2 Telemetry replay player (3D + timeline)

FRONTEND

ALIGNED

PLAN REGISTER (SOURCE A)

5 person-days · dep: G1

TIMELINE (SOURCE B)

P3-04

M (4h) · dep: P3-01

Match quality: Strong. Dependency chain agrees exactly (G2→G1 mirrors P3-04→P3-01). Effort gap (5 PD vs 4h) is the expected AI-adjusted compression for a scoped UI component, not a scope disagreement.

G3 Drone Inventory knowledge-base workflow

BOTH

ALIGNED

PLAN REGISTER (SOURCE A)

8 person-days · dep: none

TIMELINE (SOURCE B)

P4-01 – P4-05, P4-07

S/M cluster $\approx$ 1 focused day

Match quality: Strong on scope — the timeline's own note confirms "P4-01–P4-05 and P4-07 can be completed in a single focused day," matching the plan's "Both" track split. Elasticsearch (P4-06) is correctly excluded here —

it is G3's sibling gap, not G3 itself; see the note under Section 04.

G4

Analyst background worker (RabbitMQ)

BACKEND

ALIGNED

PLAN REGISTER (SOURCE A)

6 person-days · dep: none

Milestone: 21 Aug — "Analyst job worker end-to-end."

TIMELINE (SOURCE B)

P5-03

L (1.5 days) · "New infra; Docker Compose changes"

Match quality: Strong, with a naming caveat: the plan calls this RabbitMQ; the timeline implements it as Celery-with-a-broker. Both are the same infrastructure layer (Celery uses RabbitMQ/Redis as its broker), so this is consistent, not contradictory — but Action A4 in the plan evaluation ("job-state schema — PostgreSQL table vs RabbitMQ-native") should be resolved against the Celery worker directory structure the timeline assumes.

G5

YOLOv8 object-detection inference

BACKEND

WATCH

PLAN REGISTER (SOURCE A)

8 person-days · dep: G4

Director evaluation flags this as the single highest-uncertainty item and requires a pretrained-model baseline spike (Action A2, due 12 Aug) before the estimate is trusted.

TIMELINE (SOURCE B)

P5-04

L (2 days) · "ML model setup + GPU config"

Timeline's own calibration table rates ML pipeline work at only 3× speedup — the lowest multiplier alongside SITL and third-party integration — and separately lists "GPU/ML environment" under Section 11, "What AI Does Not Accelerate."

Match quality: Task exists and dependency agrees (G5→G4 mirrors P5-04→P5-03). **Watch:** the timeline's 2-day AI-adjusted figure assumes GPU/CUDA environment is already provisioned — its own Section 11 says environment setup is NOT accelerated by AI. If Action A2's baseline spike surfaces a mAP below 0.4, neither document's estimate holds, and both would need simultaneous revision, not just the plan's 8 PD figure.

G6

Video-frame extraction pipeline (ffmpeg)

BACKEND

ALIGNED

PLAN REGISTER (SOURCE A)

5 person-days · dep: G4

Plan evaluation flags ffmpeg as absent from the original tech stack document (Action A3).

TIMELINE (SOURCE B)

P5-06

M (4h) · dep: P5-03

Match quality: Strong. Both documents name it a Celery/worker-dependent task. The timeline's silence on the ffmpeg tech-stack question means Action A3 (pin version, add to Dockerfile) is still an open item regardless of task-level agreement.

G7 Change-detection module

BACKEND

ALIGNED

PLAN REGISTER (SOURCE A)

6 person-days · dep: G5, G6

TIMELINE (SOURCE B)

P5-05

L (1.5 days) · dep: P5-03

Match quality: Scope matches ("pixel-diff algorithm" = change-detection). One divergence: the plan lists G7 as depending on both G5 and G6, but the timeline's P5-05 lists its dependency as P5-03 only, not P5-04. This is likely correct as written — pixel-diff change detection does not structurally need YOLOv8 output — so the plan's dependency (via G5) may be slightly conservative rather than the timeline being wrong.

G8 MinIO object storage wiring

BACKEND

ALIGNED

PLAN REGISTER (SOURCE A)

4 person-days · dep: none

Milestone: 21 Aug, paired with G4.

TIMELINE (SOURCE B)

P5-02

S (1.5h) · "MinIO upload endpoint"

Match quality: Strong. Both pair this with the Analyst worker milestone; no dependency conflict.

G9 Analyst frontend workspace

FRONTEND

ALIGNED

PLAN REGISTER (SOURCE A)

8 person-days · dep: G4, G5

Plan evaluation Action A7 flags an open polling-interval decision (1s vs 5s).

TIMELINE (SOURCE B)

P5-07 + P5-08

M (3h) + L (1.5 days) · "job submission form" + "complex canvas rendering"

Match quality: Strong once split across the two timeline tasks — the plan's single G9 line item corresponds to both the submission form (P5-07) and the detection-results canvas viewer (P5-08). Action A7's polling-interval question is orthogonal to both and remains unresolved in either document.

G10 Automated test coverage $\geq 80\%$

BOTH

ALIGNED

PLAN REGISTER (SOURCE A)

6 person-days · dep: ongoing

Plan evaluation Action A8: no current coverage baseline documented.

TIMELINE (SOURCE B)

P6-01

L (1.5 days) · "Full pytest coverage pass"



Match quality: Strong. Note for the reader: the backend test suite has since grown substantially past this timeline's 6-June baseline (883 tests passing as of the current session) — Action A8's "run pytest-cov on Day 1" is still the correct move to get a live number before relying on either document's estimate.

G11 CI pipeline (lint, type-check, test, build)

BOTH

ALIGNED

PLAN REGISTER (SOURCE A)

4 person-days · dep: none

Plan explicitly schedules this mid-plan, once G1 and G4 exist as real code paths — a sequencing call the Director evaluation praised directly.

TIMELINE (SOURCE B)

P6-11

M (3h) · "GitHub Actions CI pipeline"



Match quality: Strong on task existence. Sequencing differs: the timeline places CI in Phase 6 (end of project), while the plan's own reasoning — and the Director's evaluation — favours running it mid-plan. This is a scheduling choice, not a scope disagreement; the plan's ordering is the one to keep.

G12 Accessibility pass (WCAG across 5 workspaces)

FRONTEND

WATCH

PLAN REGISTER (SOURCE A)

5 person-days · dep: none

Plan evaluation Action A6: WCAG level unspecified — recommends 2.1 AA as the defence-GCS baseline.

TIMELINE (SOURCE B)

P6-06

M (4h) · "Needs human visual review"



Match quality: Task exists, but the effort figures do not convert on the same basis as most other rows. The timeline's own Section 11 lists "WCAG visual review" as a fixed, human-gated bottleneck that AI does not accelerate — yet P6-06 is still coded as M (4h) rather than the ⌚ fixed-effort symbol used elsewhere in the same document for equivalent bottlenecks (e.g. SITL, GPU/ML env). The plan's 5 PD figure is more consistent with the timeline's own stated bottleneck logic than the timeline's own task-table entry is.

G13 Load testing (10+ drone telemetry + API)

BACKEND

ALIGNED

PLAN REGISTER (SOURCE A)

4 person-days · dep: G4–G8 stable

Plan evaluation Action A10: add explicit sub-100ms C2 latency target to the test plan, not just a survival check.

TIMELINE (SOURCE B)

P6-09

M (3h) · "Locust load test — 10 drones × telemetry + 3 missions"



Match quality: Strong on scope and dependency ordering (both gate this behind the full Analyst backend chain). Neither document states a latency NFR target inside the task description itself — Action A10 remains the deciding action regardless of which document's task line is used to schedule the work.

G14 Frontend automated test suite (zero→first pass)

FRONTEND

MISMATCH

PLAN REGISTER (SOURCE A)

6 person-days · dep: none

Plan milestones reference G14 twice: "31 Aug — partial" and "30 Sep — closed."

TIMELINE (SOURCE B)

— no task found

The 62-task register has no line item for a frontend (Vitest/RTL) test suite. P6-01 covers backend pytest coverage only; no equivalent frontend testing task exists anywhere across Phases 1–6.



Match quality: No corresponding task. This is a genuine omission in the timeline document, independent of the Director's plan-evaluation findings — it should be added explicitly (naturally paired with P6-01, likely inside Phase 6) rather than assumed to be folded into an existing line.

04 — FINDINGS NOT VISIBLE FROM THE GAP TABLE ALONE

Cross-reading both documents surfaces three structural points that a gap-by-gap pass alone would not show.

1 G14 has no timeline task — confirmed gap in the gap-mapping itself

Every other plan gap maps to at least one timeline task; G14 maps to none. Since the Director's evaluation already flags G10 (backend coverage) as lacking a documented baseline (Action A8), the same baseline exercise is now doubly missing on the frontend side — there is neither a plan-side DoD nor a timeline-side task to anchor it to.

2 G3 and Elasticsearch (P4-06) are correctly separate, but easy to conflate

The plan's G3 (8 PD, "none" dependency) matches the timeline's P4-01–P4-05/P4-07 cluster, which the timeline explicitly says fits in one day. The timeline's P4-06 (Elasticsearch, XL 3–4 days) is a distinct, harder task that the plan's Gap Register does not list as its own line at all — it is implicitly bundled inside G3's "Both" track. Treating G3 as an 8 PD item risks under-budgeting the Elasticsearch portion, which the timeline's own week-by-week sprint plan defers to Weeks 7–8, separate from the rest of Phase 4.

3 G12's accessibility pass is coded as acceleratable when the timeline's own bottleneck table says it isn't

This is an internal inconsistency inside the AI-Adjusted Timeline document itself, not just a mismatch against the plan: Section 11 ("What AI Does Not Accelerate") lists WCAG visual review as fixed-duration and human-gated, in the same category as SITL and GPU/ML environment setup — both of which get coded 1× (fixed) in the effort table. P6-06 is coded M (4h) instead. The plan's 5 PD figure for G12 is the more internally consistent number to carry forward.

05 — FULL CROSSWALK TABLE

Flat reference view of every row above, for copy-paste into a tracker.

GAP	DESCRIPTION	PLAN PD	PLAN DEP	TIMELINE TASK(S)	AI EFFORT
G1	3D visualisation scaffold	10	—	P3-01, P3-02	4.5–5.5d
G2	Telemetry replay player	5	G1	P3-04	4h
G3	Inventory knowledge-base workflow	8	—	P4-01–05, 07	~1d
G4	Analyst background worker	6	—	P5-03	1.5d
G5	YOLOv8 inference	8	G4	P5-04	2d
G6	Video-frame extraction (ffmpeg)	5	G4	P5-06	4h
G7	Change-detection module	6	G5, G6	P5-05	1.5d
G8	MinIO object storage wiring	4	—	P5-02	1.5h
G9	Analyst frontend workspace	8	G4, G5	P5-07, P5-08	3h + 1.5d
G10	Automated test coverage ≥80%	6	ongoing	P6-01	1.5d
G11	CI pipeline	4	—	P6-11	3h
G12	Accessibility pass (WCAG)	5	—	P6-06	4h
G13	Load testing	4	G4–G8	P6-09	3h

GAP	DESCRIPTION	PLAN PD	PLAN DEP	TIMELINE TASK(S)	AI EFFORT
G14	Frontend automated test suite	6	—	—	—

Prepared as a working cross-reference of ACSTL/DoTI/DAPE/DA/2026/001 (Director Plan Evaluation, 07 Aug 2026) against the AI-Adjusted Project Timeline (6 Jun 2026). Verdicts reflect textual agreement between the two source documents only — they do not independently re-estimate effort or re-verify implementation status against the current codebase.